Images can be created from scratch or starting from a base image. The latter is the most common approach. For example, you can start from an Ubuntu image and apply a series of modifications on top of it. The sequence of instructions would be as follows:

1. Use Ubuntu as the base image.
2. Install the Java Runtime Environment.
3. Run the `java --version` command.

Each of these instructions will generate a layer, producing the final container image shown in figure 6.2.

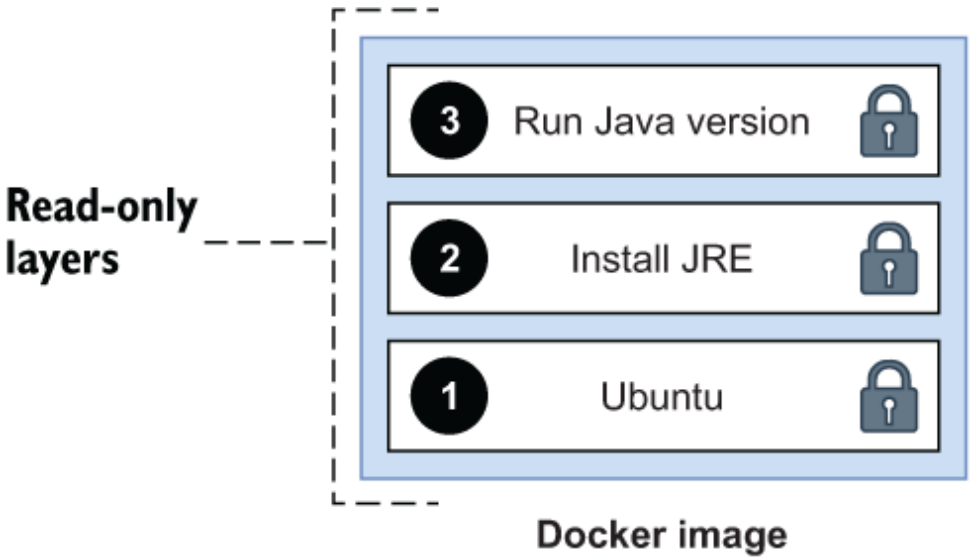


Figure 6.2 Container images are composed of an ordered sequence of read-only layers. The first one represents the base image; the others represent modifications applied on top of it.

All layers in a container image are read-only. Once they are applied, you can't modify them anymore. If you need to change something, you can do so by applying a new layer on top of it (by executing a new instruction). Changes applied to the upper layers will not affect the lower ones. This